

Freight Rate Prediction Using XGBoost Regression

Technical Assessment Report | Machine Learning Engineer Candidate

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Report Type	Freight Rate Prediction — Technical Assessment
Model Family	XGBoost Regressor

Objective

This report documents the development of a machine learning regression model designed to accurately predict freight load rates using historical shipment data. The model was trained to generate predictions for a held-out validation set as well as a fixed December 2025 scenario, providing a reference benchmark for evaluating pricing behavior over a defined time horizon.

Dataset Summary

Item	Value
Source Dataset	train-test.csv
Total Records	48,000
Validation Dataset	validation.csv
December Scenario	31 prediction rows
Target Variable	posted_rate
Problem Type	Supervised Regression

Business Problem

The objective is to estimate the posted freight rate for individual shipments using operational, geographical, and market-related features, including pickup location, delivery location, equipment type, shipment weight, travel distance, and prevailing market conditions. An accurate rate estimate supports more consistent pricing decisions and reduces reliance on manual, experience-based rate quoting.

Data Validation & Modeling Approach

Data Quality Assessment

Exploratory analysis of the training dataset surfaced a small number of data quality issues, each addressed prior to model training to preserve the integrity of downstream feature engineering.

Issue Identified	Resolution Applied
300 missing weight values	Median imputation
374 missing market_index values	Median imputation
Negative shipment weights	Converted to absolute values
Date stored as string	Converted to datetime; temporal features engineered

Feature Engineering

A set of derived features was engineered to expose temporal, geographic, and operational signal not directly present in the raw shipment records:

Engineered Feature	Purpose
Year / Month / Day	Capture temporal pricing trends
Day of Week	Weekly demand patterns
Haversine Distance	Geographic travel estimation
Weight per Mile	Load efficiency indicator
Market Distance	Interaction between demand and distance
Pickup & Delivery State	Regional freight behavior

Training & Validation Strategy

The original **train-test.csv** dataset containing 48,000 shipment records was internally divided using an **80:20 train-validation split** with **random_state = 42** to ensure reproducibility. Model optimization was performed using **RandomizedSearchCV** with **5-fold cross-validation**, evaluating **12 hyperparameter combinations**. The final XGBoost configuration was selected based on the lowest cross-validation RMSE before being evaluated on the held-out validation dataset.

Model Selection

The XGBoost Regressor was selected as the final modeling approach. Gradient-boosted tree ensembles of this type perform exceptionally well on structured, tabular regression problems and are well suited to modeling the non-linear relationships that exist between operational and market-driven pricing features in freight data.

Results & Conclusion

Model Performance

Metric	Score
R ² Score	0.8613
RMSE	544.68
MAE	126.32
Best Cross-Validation RMSE	622.79

The optimized XGBoost model explains **86.13%** of the variance in freight pricing while maintaining an average prediction error of approximately **\$126**, demonstrating strong predictive performance on unseen validation data.

December 2025 Fixed Prediction

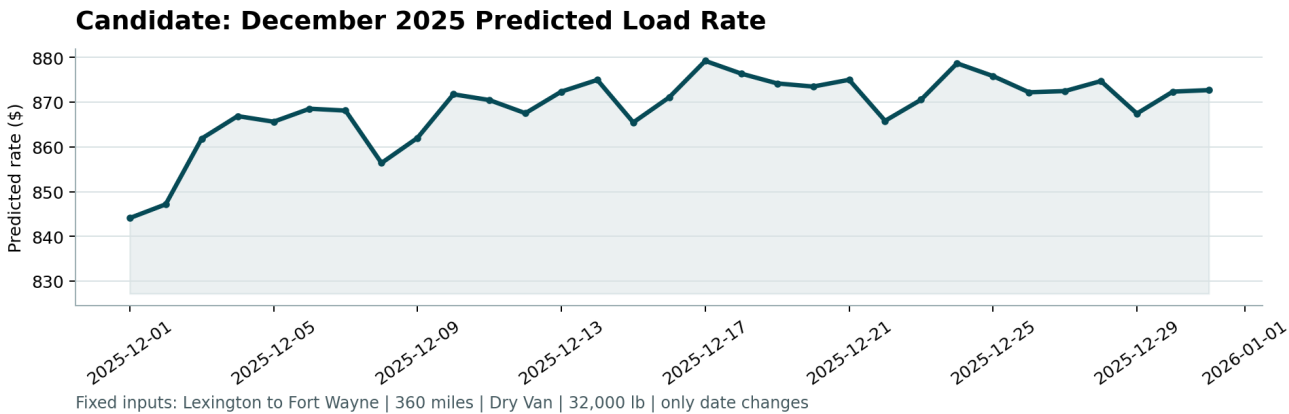


Figure 1. Predicted freight rates for the fixed Lexington → Fort Wayne route during December 2025.

Conclusion

- Successfully developed an end-to-end freight rate prediction pipeline, from raw data ingestion through final rate estimation.
- Implemented robust preprocessing and feature engineering to improve model accuracy and stability.
- Optimized the XGBoost model using cross-validated hyperparameter tuning to maximize generalization performance.
- Generated submission-ready validation and December prediction outputs, validated using the provided scoring utility.